MAZRA'AT ALBAAN

Project Vuna

Credit, certified inputs, and parametric drought insurance — bundled and delivered to South African smallholder farmers through their cooperatives. On Solana.

HACKATHON

Solana 2026 Frontier — Physical World Applications

AUTHOR

Tumo Mogame

REPOSITORY

github.com/TUMO-MOGAME/Solana-Based-Agricultural-Marketplace

solana-based-agricultural-marketpla.vercel.app

The problem

Africa's smallholders feed the continent — and they do it on the thinnest financial margin of any farmer category in the world.

70%
of Sub-Saharan Africa's food is grown by smallholders

<5%
of commercial bank lending reaches them

<3%
of crops are insured against weather risk

Five real things going wrong

- bullet **Credit is locked behind collateral they don't have.** No title deed, no nearby branch, no formal payslip means no loan — even when they've farmed the same land for thirty years.
- bullet **Inputs are unaffordable, imported, and often counterfeit.** A farmer who pays for fake seed loses the entire season.
- bullet **One drought wipes out a family.** There's no buffer — savings, insurance, or otherwise. Recovery takes years that the next season doesn't grant.
- bullet **Middlemen capture 40–60% of crop value.** Cash-on-delivery quotes at the gate are deliberately low. The farmer accepts because the alternative is the crop rotting.
- bullet **Good farmers can't prove they're good.** Every season starts from zero. Lenders have no way to tell a reliable repayer from a first-time defaulter.

The Grow Pack

Our product is a single bundled offer farmers actually want to take: **credit + certified inputs + parametric drought cover**, repaid at harvest, sold through their cooperative.

How a season looks for a farmer named Sipho

1. Sipho registers at his cooperative. The co-op vouches he is who he says he is.
2. He **applies for a Grow Pack** through the app: 2 hectares of maize, drought cover, total around R 1,650.
3. The cooperative approves it. **Suppliers deliver seed and fertilizer directly** to the depot. Sipho never handles cash for the inputs.
4. **Drought cover goes live**. If rainfall in his area drops below an agreed threshold, the policy pays out automatically — no claim form, no investigator visit.
5. At harvest, Sipho sells his maize. **The bundle cost plus a small service fee is auto-deducted** from the proceeds. He keeps the rest.
6. His **credit history is portable**. Next season's offer is better because this season's was repaid.

Three rules we never break

bullet **Hide the chain.** Sipho never sees "blockchain", "wallet", "USDC", or "Solana". He sees Rand and his own language.

bullet **Mobile-first.** Built for a R 800 Android, often offline. Voice support for low-literacy users.

bullet **Cooperative as referee.** The trust isn't in the code — it's in the partner organisation Sipho already knows.

What we've shipped

Phase 1 is a working devnet demo with end-to-end integration between the on-chain program and a real farmer-facing dashboard.

On-chain (Solana / Anchor / Rust)

- **Six instructions** covering the full Grow Pack lifecycle: register_farmer, request_grow_pack, approve, disburse, trigger_insurance_payout, settle_repayment.
- **99 TypeScript tests + 44 Rust tests** (143 total) covering credit-score logic, pricing, parametric trigger tiers, harvest-settlement math, and the full end-to-end lifecycle in an in-process litesvm runtime.
- **Deployed to Solana devnet.** Real PDAs, real transactions, real on-chain reads from the dashboard.

Farmer dashboard (Next.js, deployed on Vercel)

- **Active** tab — current Grow Pack status, weather, credit history, voice-narrated daily summary.
- **Apply** tab — submit a Grow Pack request directly to devnet, signed by the connected wallet. Pre-flight check prevents duplicate seasonal packs.
- **Insurance** tab — drought-payout banner with rainfall tier visualisation. Voice-narrated payout announcement.
- **History** tab — past seasons (last three years) read from the chain, status pills, jump to detailed view.
- **Marketplace** tab — Phase 1 UI showing direct buyer offers (mills, retailers, brewers) with explicit +X%-vs-middleman comparison. Phase 2 escrow logic outlined in this document.
- **Wallet** connect for Phantom and Solflare via wallet-adapter-react. Persisted via Wallet Standard.

Accessibility

- **Voice playback (ElevenLabs).** Active-tab summary and drought-payout banner read aloud — designed for low-literacy farmers who'd rather hear than read. Streams progressively so playback starts within ~2 seconds.
- **Hide-the-chain.** Currency is Rand throughout. No wallet jargon in any farmer-facing copy.

Phase 2 — On-chain marketplace escrow

Phase 1 shows direct-buyer prices. Phase 2 makes those matches **actually move money** safely between buyer and farmer.

The story

Sipho sees that Lebone Mills is buying maize at R 5,400/ton — around 28% more than the local middleman pays. He taps **Match buyer** for 2 tons.

Lebone Mills receives the match request and **locks R 10,800** (2 tons × R 5,400) into a holding pool that neither side can pull back unilaterally. Sipho can see the funds are committed.

Sipho harvests, delivers to the mill. The cooperative officer opens the co-op app and confirms: **"Yes, Sipho delivered 2 tons of maize on October 12."**

That confirmation triggers automatic release. **R 10,800 lands in Sipho's account within minutes** — not 30 days, not 60 days, not after a phone call begging for payment.

What this fixes — for both sides

- **Sipho can't be scammed at the gate.** The price is locked the moment the buyer commits. No "actually I'll pay R 4,200" at the truck.
- **Lebone Mills can't be scammed either.** Funds only move after the cooperative confirms the maize arrived.
- **The cooperative becomes a referee, not a collector.** They already know Sipho. The app turns their trust into a single tap that moves real money.
- **No 30-90 day payment lag.** Working capital, on time. That alone changes whether a smallholder survives the next input cycle.

Governance — graduated trust

Code that holds money has to be honest about who can change the rules and when. Our plan tightens control as real money starts flowing — never the other way around.

Three stages of control

Stage	Who can change rules	Why this is right
Hackathon (now)	Authors only — single signing key	No real money at stake. Speed of iteration matters more than rule st
Pilot v1 (50–200 farmers, 2027)	Multisig: at least 2 of [authors, auditor, cooperative leader]	Real money is starting to flow. No one party can change rules unilat
Production (regional, 2028+)	DAO: farmers, cooperatives, and partners vote on changes	Trust scales with the user base. Eventually, no one has unilateral up

The two non-negotiables

bullet **Old deals always keep old rules.** If a buyer locks R 10,000 under the rule "release in 30 days", and we later change the default to 45, the original R 10,000 still releases on day 30. New rules apply only to *new* commitments.

bullet **Rule changes are announced, never silent.** Every farmer and buyer using the platform sees what changed before the change goes live. Quiet upgrades are how trust dies.

Why this matters

Smallholder farmers have been burned by unilateral changes before — by lenders who hike rates mid-loan, by buyers who haggle down at the gate, by regulators who change subsidies without notice. The graduated-trust governance is the part of Phase 2 that says *we're not building another tool that can be turned against you*. The code has to prove it.

Roadmap

Slow, careful, honest scaling. Each step proves a hypothesis before the next one is funded.

Phase	Window	Goal
Hackathon	6 weeks (now)	Working devnet demo, end-to-end. 5–10 simulated farmers.
Pilot v1	H1 2027	50–200 real farmers, one province, one crop. Target <b>60–80% repaym
Pilot v2	H2 2027	Second province, second crop. Reinsurance partnership begins. Phase 2 escrow live with r
Regional	2028+	Zambia and Kenya through Superteam Africa. Multi-currency settlement. DAO governance

What we are deliberately NOT building

- bullet **A supply chain** — we partner with existing certified-seed and fertilizer suppliers.
- bullet **Logistics** — suppliers fulfil through their own channels.
- bullet **An underwriting fund** — we pair with a licensed insurer or DeFi pool. We are not the balance sheet.
- bullet **A new payments rail** — settlement is in stablecoin, then reconciled to Rand via existing on/off-ramps and PayShap.

Failure modes we plan against, not around

- bullet Cooperatives never sign on → outreach starts week one, not month six.
- bullet Repayment crashes below 50% → conservative loan size, group lending fallback.
- bullet Climate wipes out the pool → reinsurance partner before it grows beyond R 1m.
- bullet Regulatory action → no retail solicitation pre-licence.
- bullet Smart-contract exploit → audit before mainnet, bounty programme.

Tech stack

Layer	Tool	Why this one
Blockchain	Solana (devnet now, mainnet for pilot)	Sub-cent fees, sub-second finality, the right ecosystem for African mobile-first
Smart contracts	Anchor (Rust)	Standard Solana framework, audit-friendly, mature toolchain.
Oracle	TBD (alternative to Pyth)	Pyth has no weather data — we are evaluating Switchboard custom fee
Frontend	Next.js + Tailwind (App Router, React 19)	Fast to ship, good wallet adapters, Vercel deployment.
Wallet	Phantom / Solflare (now), Magic.link or Privy (production)	Custodial UX so the farmer never sees a seed phrase.
Voice	ElevenLabs (Flash v2.5)	Low-latency multilingual TTS; we are auditing isiZulu and isiXhosa quality.
Backend / DB	Supabase (Postgres + Auth)	Fastest path to working auth + KYC storage. Replace with proprietary stack if
Hosting	Vercel + Supabase	Cheap, fast, low ops — right for early stage.

Regulatory — South Africa

These are non-negotiable. We are designing for compliance from day one, not retrofitting it after a headline.

- bullet **National Credit Act** — NCR registration to lend.
- bullet **FAIS** — FSP licence to sell insurance product.
- bullet **FSCA / CASP framework** — licence to handle crypto.
- bullet **SARB Exchange Control 2026** — approval before any cross-border crypto flow.
- bullet **POPIA** — strict farmer-data handling. R 10m fine on breach.
- bullet **Insurance Act 2017** — parametric insurance still requires a licensed underwriter.

We hold these licences ourselves or partner with a holder. There are no shortcuts.

What we're looking for

This stage of the project is partnership-bound. The hardest problems are off-chain — trust, distribution, and licence coverage — and we want partners on each.

Three asks

- bullet **A licensed agri-insurer** willing to underwrite the parametric drought policy embedded in each Grow Pack. We bring distribution, onboarding, data, and automated payout execution. They bring the licence and the balance sheet. We have a draft product brief ready.
- bullet **A cooperative pilot partner** — one province, one crop, 50–200 farmers. We provide the technology, training, and support. The cooperative provides KYC, trust, and the human layer we cannot automate.
- bullet **An audit partner** for the on-chain code before any real Rand flows. Phase 2 escrow goes nowhere near mainnet without it.

Closing thought

> *Smallholders grow most of the food in our country and on our continent. They get under five percent of the credit. The gap is not a market failure — it's a distribution and trust failure. Mazra'at albaan closes a slice of it with software that the farmer never has to understand to benefit from.*

Contact

Tumo Mogame

emma.m.strategy@gmail.com

github.com/TUMO-MOGAME/Solana-Based-Agricultural-Marketplace

solana-based-agricultural-marketpla.vercel.app